

ECONOMIC EVALUATION AND ANALYSIS OF THE GERMAN AND FRENCH GRIDS ENVIRONMENTAL FOOTPRINT WITHIN THE SCOPE OF PRODUCTION AND IMPLICATIONS OF NUCLEAR POWER PLANT CLOSURES

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Abstract

The environmental footprint of electricity generation has very often focused on the carbon footprint aspect alone. However, greenhouse gas emissions are not the only source of anthropogenic pressure over the environment (human or non-human). Acidification, eutrophication, ionizing radiation, the use of resources and water, etc. all contribute to the overall impact of our activities, notably electricity generation. Life-cycle analysis (LCA) is a robust quantification methodology of material and energy flows resulting from the production of a kWh of electricity within a geographical area. To ensure that these results are operational for a policy maker, we use weighting methods, which aggregate a list of impacts into a single score. Monetary valuation is one way of doing so, whereby the score is expressed in monetary value. Based on a review of the literature on electricity generation LCA and monetary valuations, we first discuss methodological principles. Then, using hourly electricity production time series by technology in France and Germany, we calculate an environmental cost of electricity production on an annual average. The results suggest that, for an electricity system in transition, there is a strong economic rationale, if environmental costs were internalized, to keep nuclear capacity in service and/or to guarantee the best operational performance of these plants. This article is the first part of a broader research question, that is investigate the marginal impact of increasing or decreasing power output of various technologies over the environmental performance (cost) of power systems. Thus, next step is to implement an econometric model on these hourly cost and electricity output chronicles, to draw quantitative conclusions regarding this rationale.

Preliminary Remarks

This paper is the result of an ongoing research and as such will focus on motivation, methodological considerations, and the first results, which provide material for further work. Hence, results included are not definitive and some aspects of this work will not be mentioned in this paper, most notably those regarding the econometric model which assumptions and validity is still in discussion.

1. INTRODUCTION

Broadly speaking, environmental issues (Human health, Natural environment, and Natural resources as endpoint indicator, in academic usage, see below) have a growing place on the political, scientific, economic, media and cultural scene. For businesses and academics, the benchmark methodology for assessing the impact of products and services is life cycle assessment (LCA) [1-2]. LCA makes it possible to monitor the environmental impacts of a wide range of products and services "from cradle to grave". More specifically, we can frame the various impacts resulting from an inventory of material and energy flows over the life of the product, in the form of intermediate indicators or intermediate measurements. The European Commission has developed a standardised framework called the 'environmental footprint' (EF), which we will use in this work [3]. Although LCA does not cover all the goals set by sustainable development framework adopted by United Nations in 2015, and as such deserves to be "extended" [4], LCA is still recognised as the most widely used tool for environmental policy and plays a central role in addressing the sustainability of production and consumption. Electricity generation is both a key driver of environmental impact - it is responsible for 42% of global CO₂ emissions in 2022 [5] - and has a crucial role to play in addressing these issues - the shift to low-carbon electric power systems (so-called electrification) is crucial to achieve net zero emissions by 2050 for the global energy sector [6]. Therefore, LCA has been widely used to assess electricity generation, with a growing number of studies in the academic literature [7].

Two conclusions can be drawn from academic literature [7]: firstly, there is a geographical bias, with most studies conducted in Europe; secondly, climate change is the most investigated midpoint indicator. The former

aspect is not an issue for the present work as we only consider electricity generation for European countries, namely France and Germany, but should be carefully considered for global assessment as industrial processes, environmental standards, and regulations, differ widely between countries. The latter is more of concern since we should refer to all the 16 midpoint indicators of EF in this work. In LCA literature, studies assessing those indicators for various generation technologies are few.

Interpreting the results of an LCA from indicators which encompass complex physicochemical and biological processes, and drawing relevant regulations from it might be a challenging task to ask policy makers. Hence, from a pragmatic point of view, there is a need for translation and aggregation into a unique metric. All quantitative valuation methods imply a weighting procedure [8]. The ISO 14044 standard defines weighting as “converting and possibly aggregating indicator results across impact categories using numerical factors based on value-choices” [9]. It also recognizes weighting as “not scientifically based”, where science is limited here to natural sciences. To the very least, it is therefore crucial to discuss and clearly identify political, ideological, and ethical values which are at stake and their methodological translation (e.g., discount rate) [8]. Monetary valuation is one of the mostly applied ways of weighting.

From economists’ point of view, monetary weighting (or valuation) methods objectify the parameters used within public environmental policies. Those are required when external (positive or negative) effects, so called “externalities”, are not mediated by economic institutions (markets or regulations in liberalised economies) [10]. In such conditions, actual value of a product is not fully reflected¹, something that causes market failures. A sub-optimal allocation of resources follows according to welfare economics, resulting in a sub-optimal overall well-being of utility maximisers agents. Taxes (e.g., in €/MWh for electricity generation) might be implemented as a market-based regulatory tool to bridge the gap between social cost and private cost [11], the former being greater than the latter in the case of negative externalities, that is to say, when private productive activities give rise to social losses of various kinds. Economists should be very careful when setting such taxes [12] particularly with the risk of double counting for effects that are already partially mediated through markets [13] (e.g., ionizing radiation for civil nuclear workers or carbon emission with the EU Emissions Trading System). Other scholars argue that “social costs are implicit in the competitive structure of contemporary society and are merely the by-product of any process of rapid economic change and growth” [14] - in a nutshell, a temporary price to pay for longer prosperity.

In practice there are two basic methodological approaches used for the valuation of externality impacts: the abatement cost approach and the damage cost approach [15]. The abatement cost approach takes the cost of regulations to mitigate and control damage generated by the externality, as a proxy for the value of avoided damage. The major drawback in this approach is that it relies entirely on the ability of policy makers to make the right assessments. Often, it underestimates those costs [10, 15]. The damage cost approach can be further divided into two approaches: top-down and bottom-up. Top-down approaches attribute global damage costs to various global emissions of pollutants to evaluate the external cost per unit of emission (e.g., what is the cost of national healthcare related to respiratory disease incurred by national particulate matter emissions inventory?). Consequently, damages not valued means damages ignored (and thus, potentially not internalised). Bottom-up approach depends on whether there is an observable market to assess the cost of a damage. If there is, market prices provide the desired valuation. If not, we can proceed either through direct or indirect methods; direct methods simulate markets involving surveys, experimental processes etc. Contingent² valuation or stated preference methods are among the most common and the most criticized outside (see discussion below) and inside practitioners’ sphere [16, 17]. In framing and designing a study, whether the total economic value (see footnote 2) or a subset should be assessed is mandatory [16]; indirect methods which infer value through agents’ behavior changes on markets resulting from changes of various kinds (e.g., valuing noise level through housing rents spatial analysis), hence the name “revealed preferences”. Bottom approach is generally site specific, and evaluation are in principle not transferable. However, following cautious guidelines [18] so called “transfer methods” allow for such transfers [16,19-20]. Institutional approaches [21], rejecting the reduction of (economic) institutions (such as markets) to individual preferences, exist but are not as operational and elaborated as neoclassical approaches [16].

¹ Total economic value encompasses “usage value” (whether direct, indirect or as an option in the future) and “non-usage value” (inheritance value or existence value) [10]. From a philosophical stand view, values are commonly classified as instrumental or intrinsic. Broadly speaking, instrumental values correspond to direct, indirect, option and inheritance values, while existence value might reflect intrinsic value.

² Named after the fact that the elicited values are contingent upon the markets simulated by the respondents [22].

Many critics have been addressed to monetary valuation methods: its dependency on a cultural, economic, sociological and political contexts and its incorporate of implicit ethical views [22-24]; the extension of jurisdictional authority over unlimited set of unmeasurable objects whether human related or not (“we need to price nature to protect it”) [22-23]; the Arrow impossibility theorem should prevent from individual preferences aggregation in contingent valuation [21]; the huge disparities between “Willingness to pay” (WTP) and “Willing to accept” (WTA) in valuing some non-market goods (with a ratio ranging from 1,6 to 6,5) [16, 26]; lack of information, understanding, and therefore the irrelevance of assessments over the environmental impacts involved [16]; structure of individuals’ rationality implied in those models [23] ; unreasonable assumptions [27]; incomes dependency of surveys participants [16, 28] etc.

Discussion of those highly relevant and critical issues is beyond the scope of this paper. Nonetheless, we shall outline the arguments which, in our view, make it appropriate to use monetary valuation in this work. First, the moment we perform a full LCA, it might be argued that a weighting procedure has been accepted implicitly as, should some aspects of the environment be unvaluable, we will already address those aspects before performing the LCA [8]. Second, both individuals and societies already assess the value of incommensurable objects (health, ecosystem etc.) in our everyday and collective decision-making [29]. To put a value tag on such objects, beyond the obvious strong anthropocentrism gesture it symbolizes, can be beneficial for these objects [30]. Third, using those valuation techniques do not exclude the possibility (or the necessity) to discuss and elaborate more accurate methodologies [31], or even to include moral philosophers into the process. Overall, LCA practitioners view weighting as relevant for decision marking but insist on the need to improve uncertainty and robustness [31]. Forth, using the “measuring Rod of Money”, whose limits are well understood by most economists [32], provide deep insights when comparing several options, particularly regarding the choice of electricity generation mix where broad conclusions, such as key variables underlying carbon intensity or rough comparison of environmental costs of electricity generation between two different mixes can be drawn. Fifth and finally, we should not confuse the monetary valuation of X with the true value (whatever the sense we give to the concept) of X [33]. Monetary valuation is merely a point of view from which a decision can be objectivized, in the sense that it is not left to the discretion of politicians (they are given less power in the decision-making) but is the result of a calculation on the terms of which a relative methodological consensus can be reached.

2. MATERIALS AND METHODS

Electricity mix and its environmental impacts vary both in the short and in the long term. Although important progress has been made, very few studies address with high temporal resolution (hours) those impacts [34-35]. In this study, we conduct an environmental cost assessment of the hourly power generation by technologies for the French and the German grids from data ranging from 2018 to 2021 provided by Electricity Maps³. We do not consider the electricity price provided for two reasons: the first being the difficulty to aggregate prices from different time horizons markets; the second being the double-counting errors (see above). Therefore, we will only consider environmental cost stemming from the kWh monetary valuation of generation technologies. We used national statistics [36-39] to set out breakdown by technology (Poly-Si, CdTe, CIGS) for solar generation, as well as assumptions regarding ground-mounted vs. roof-mounted based on the nominal power of installations. Same goes for offshore and onshore wind generation in Germany with a corrective coefficient accounting for the load factor differential to assess accurate hourly outputs.

Valuations are derived from an extensive review of literature of monetary valuations corresponding to one or more midpoint indicators framed in EF [40]. Methods applied are specified. When available (for 13 midpoints indicator over 16), we chose to keep only valuation coefficients assessed through top-down damage approach as it is less subject to controversies and the most common approach used in [40]. Otherwise, we kept coefficients drawn from market price methods. For calculation we used median value after proper unit conversions as “the degree of uncertainty affecting extreme values, particularly high values, is transferred to the mean but does not affect the median. From this point of view, the median is a more reliable indicator » (INSEE, French institute of Statistics). For particulates, we adopted a consolidate result from literature [41] stating that the particulate matter < 2.5 µm (PM2.5) emissions dominate the overall characterization factor of PM10 emissions, in other words that PM2.5

³ <https://app.electricitymaps.com/map>

fractions account for almost all the damage incurred by PM10 emissions. Table 1 summarizes data collected (hereafter, prices are expressed in 2019 euros by default).

TABLE 1. LIST OF MONETARY VALUATION COEFFICIENTS IN €₂₀₁₉. (NB: Details can be found in [40]. “Respiratory effects, inorganics” and “particulate matter” refer to the same indicator expressed in two different units. We used both in this work (see below))

Method	Midpoint impact indicator	Median value	Number of valuations included
Damage cost	Climate change (€/kg eq. CO ₂)	1,86E-01	4
Damage cost	Acidification (€/mol eq. H ⁺)	1,03E+00	7
Damage cost	Freshwater ecotoxicity (€/CTU)	3,88E-05	2
Damage cost	Freshwater eutrophication (€/kg eq. P)	3,83E+00	6
Damage cost	Marine eutrophication (€/kg eq. N)	1,61E+00	4
Damage cost	Terrestrial eutrophication (€/mol eq. N)	1,12E-01	3
Damage cost	Carcinogenic effects (€/CTUh)	5,75E+05	5
Damage cost	Ionising radiation (€/kg eq. U235)	1,22E-03	5
Damage cost	Non-carcinogenic effects (€/CTUh)	5,75E+05	5
Damage cost	Ozone layer depletion (€/kg eq. CFC-11)	4,61E+01	5
Damage cost	Photochemical ozone creation (€/kg eq. NMVOC)	1,18E+00	7
Damage cost	Respiratory effects, inorganics (€/disease)	7,98E+05	1
Market price	Particulate matter (€/kg eq. PM _{2.5})	3,36E+01	4
Market price	Dissipated water (€/m ³)	1,43E-02	2
Damage cost	Fossils depletion (€/MJ)	1,03E-02	8
Damage cost	Land use (€/pts)	2,03E-03	2
Market price	Mineral and metals depletion (€/kg eq. Sb)	1,92E+01	6

LCA for generation technologies used in this work is performed at European level [42] for all the impact indicators framed in EF (see Tab. 1.). For “respiratory effects, inorganics” indicators provided in [42], we compared statistically results with existing literature [43-45]. We identified significant differences in monetary valuation for two technologies – natural gas and nuclear - which we addressed using other studies assessing the impact in terms of particulate matter emissions [35 (supplementary materials), 43 (supplementary materials), 44]. As no inventory for lignite (brown coal) was provided in [42], we aggregated impacts from brown and hard coal into hard coal, which is a favourable assumption [46]. Without hard coal technological breakdown, we assumed an average impact for each midpoint indicator.

Regarding nuclear impacts⁴, we compared results with [44] (Tab. 2). Major differences were found over freshwater eutrophication indicator discussion of which is beyond the scope of this paper. Furthermore, this difference were second order matter when monetary valuation was performed. Differences in electricity origin for conversion and enrichment might explain partially those discrepancies (Fig. 1.) as electricity used during those processes is comparatively more decarbonized in [44] assumptions. Another explanation may stem from the fact that [42] did not consider fuel recycling in its assumptions leading to, all things being equal, extra mining - which is diesel and carbon-based electricity intensive - compared to [44]. There is an unresolved issue around ionizing

⁴ We chose to keep pessimistic value for “ionising radiation” indicator.

radiation impact: in [44] they are dominated by Carbone 14 (C14) emissions during production specifically, whereas in [42] they are dominated by Radon 222 emissions during mining stage, maybe due to a failure (or a lack of) to account for C14 emissions. In turn, this leads to an underestimation of several order in ionizing radiation impact compared to [44]. Fortunately [42] also provides a “pessimistic” value, similar in magnitude to the one in [44]. We chose to keep the latter for the calculations. Note that the “water withdrawal” impact indicator covers not only water evaporated into the atmosphere via cooling towers, which range around 2 liters/kWh for nuclear energy based on empirical analysis performed on the French nuclear fleet [47], but also water returned to the initial environment (rivers etc.) for open-circuit thermal power plants. There is thus a catch in using market price for water consumption when in the case of thermal power plant, most of the water is restored immediately after cooling. We will proceed as such, however.

FIG. 1. Comparison of electricity mixes for conversion and enrichment used in [42] and [44] (author's calculations based on various sources, including [44] for supply portfolio data for France, and national statistics for electricity mix)

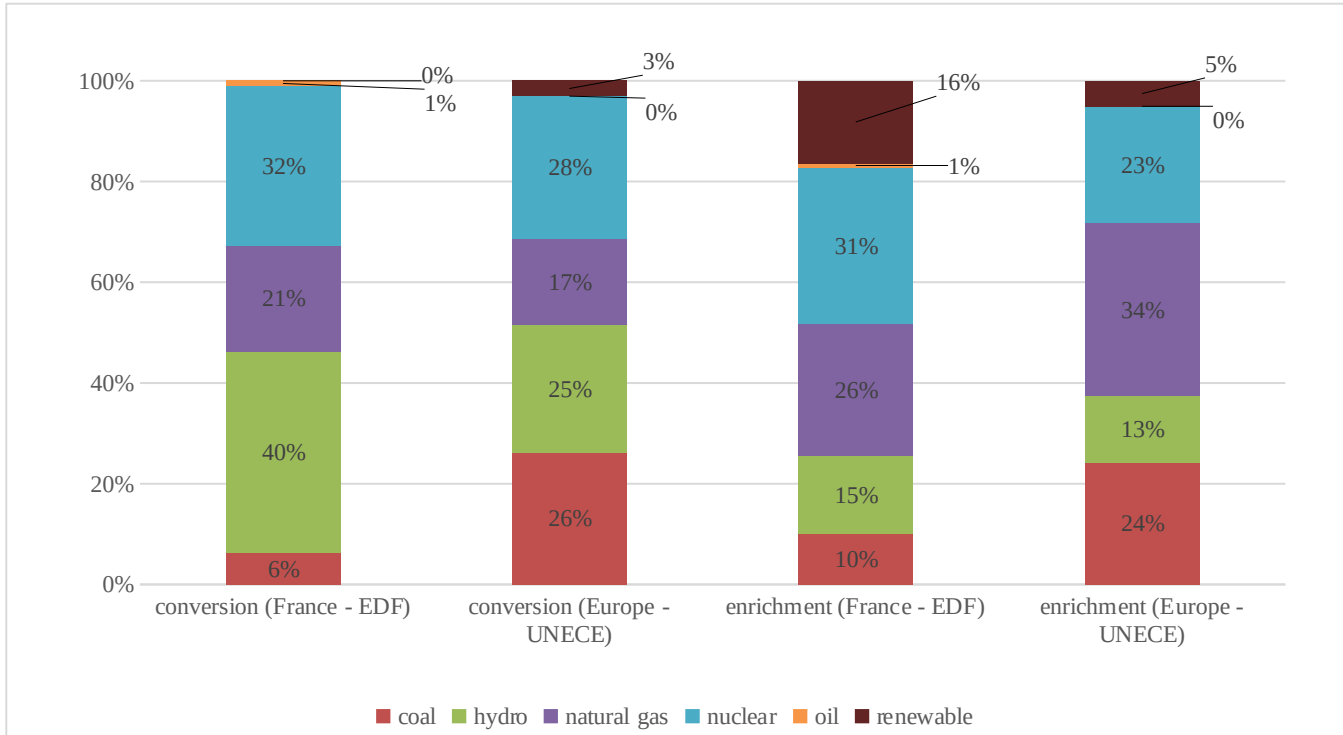


TABLE 2. COMPARISON BETWEEN TWO LCA OF ONE NUCLEAR KWH GENERATED [42, 44] OVER SEVERAL IMPACT MIDPOINT INDICATORS (Indication: CC: climate change, O: ozone layer depletion, IR: ionizing radiation, POC: photochemical ozone creation, A: acidification, FE: freshwater eutrophication, TE: terrestrial eutrophication, ME: marine eutrophication, MMD: minerals and metals depletion)

Indicator (emissions per kWh produced)	CC (kg eq. CO ₂)	O (kg eq. CFC-11)	IR (kg eq. U235)	POC (kg eq. NMVOC)	A (mol eq. H ⁺)	FE (kg eq. P)	TE (mol eq. N)	ME (kg eq. N)	MMD (kg eq. Sb)
[44]	3,7e-3	4,7e-10	6,80e-1	2,10e-5	3,40e-5	1,90e-7	7,10e-5	1,50e-5	4,80e-6
[42]	5,29e-3	4,62e-10	4,75e-1	2,65e-5	4,28e-5	6,45e-6	9,70e-5	8,20e-5	3,33e-7

A critical issue identified in [42], is that in the most recent “Product Environmental Footprint Category Rules” guidelines (PEFCR) [48], uranium is accounted as “fossil” in the methodology, which is “technically not correct” [42]. Moreover, in the study which grounds this aspect of the methodology in PEFCR [49], uranium is part

of the impact category ‘Mineral and metals depletion’ since “uranium is used for many different applications, not only for energy supply” [50]. [42] gives a figure both including (16,4 MJ/kWh) and excluding (0,07 MJ/kWh) uranium in fossil depletions category. In line with academic literature [44, 50], we have chosen to include it in “Mineral and metals depletion” category accounting for its non-renewable⁵ but also non-fossil character. We used most recent abiotic depletion potential assessment of uranium [50] (as characterisation factor for the depletion category), which adds $2,82 \times 10^{-7}$ kg eq. Sb/kWh (assuming 0,20-0,30 mg Uore/kWh [42, 44]) to the initial “Mineral and metals depletion” impact evaluation ($3,33 \times 10^{-7}$ kg eq. Sb/kWh). Finally, despite significant differences between [42] and [44] on this indicator, which have not been further investigated, the impact on environmental costs is low (+/- 0,10 €/MWh).

Hydro impacts were calculated from two inventories [42]. Several indicators vary by several orders of magnitude. Indeed, one has very conservative assumptions [43 (supplementary materials)] and provide highly skewed results when compared with existing literature of hydro impacts in western Europe [45, 51, 52]. We thus choose to keep results from the most favorable one. Finally, as no LCA is performed for oil power plant in [42], we arbitrarily choose an environmental cost that reflects an intermediate environmental in impact coal and gas [35 (supplementary materials)]. Sensibility studies carried on showed that the oil cost assumption has a minor impact on the French environmental cost of electricity generation as its share in overall power output is limited (Fig. 2.). However, we did not proceed as such for biomass or other renewable energies, leaving them aside of the analysis. Consequently, they were not included in environmental costs computation.

3. PRELIMINARY RESULTS

Table 3 gives the environmental cost in €/MWh as calculated following the assumptions and the calculations presented above (without CIGS technology⁶ as it is irrelevant for the analysis). Hydro has the lowest environmental cost (6 €/MWh), followed by nuclear (9 €/MWh), and onshore/offshore wind (respectively 11 and 12 €/MWh). Among low-carbon technologies, polysilicon photovoltaic (PV) has the highest cost (28 €/MWh). However, what matters most is not the exact environmental costing of the technologies, but the order of magnitude and the ranking, what we may call the “environmental merit order” following our central hypothesis that all the impacts are borne solely by the production of one unit of electricity. In this regard, we observe a net gap between fossil-based technologies and non-fossil-based technologies, or low-carbon technologies: thus observing a very steep environmental merit order curve. For the latter, the cost is typically around 10-20 €/MWh whereas for the former, the costs exceed a hundred euros per MWh. As expected from LCA of carbon-based technologies [51], natural gas has the lowest cost and hard coal is, by far, the most environmentally costly technology.

TABLE 3. Environmental costs of power generation technology by merit order

Power generation technology	Environmental cost (€/MWh)	Costs
Hydro	6	
Nuclear	9	
Onshore wind	11	
PV CdTe	12	
Offshore wind	12	
PV CdTe, ground-mounted	12	
PV poly-Si, roof-mounted	28	
PV poly-Si, ground-mounted	28	
Nature Gas Combined Cycle	170	
Oil	250	
Hard coal (and brown coal)	400	

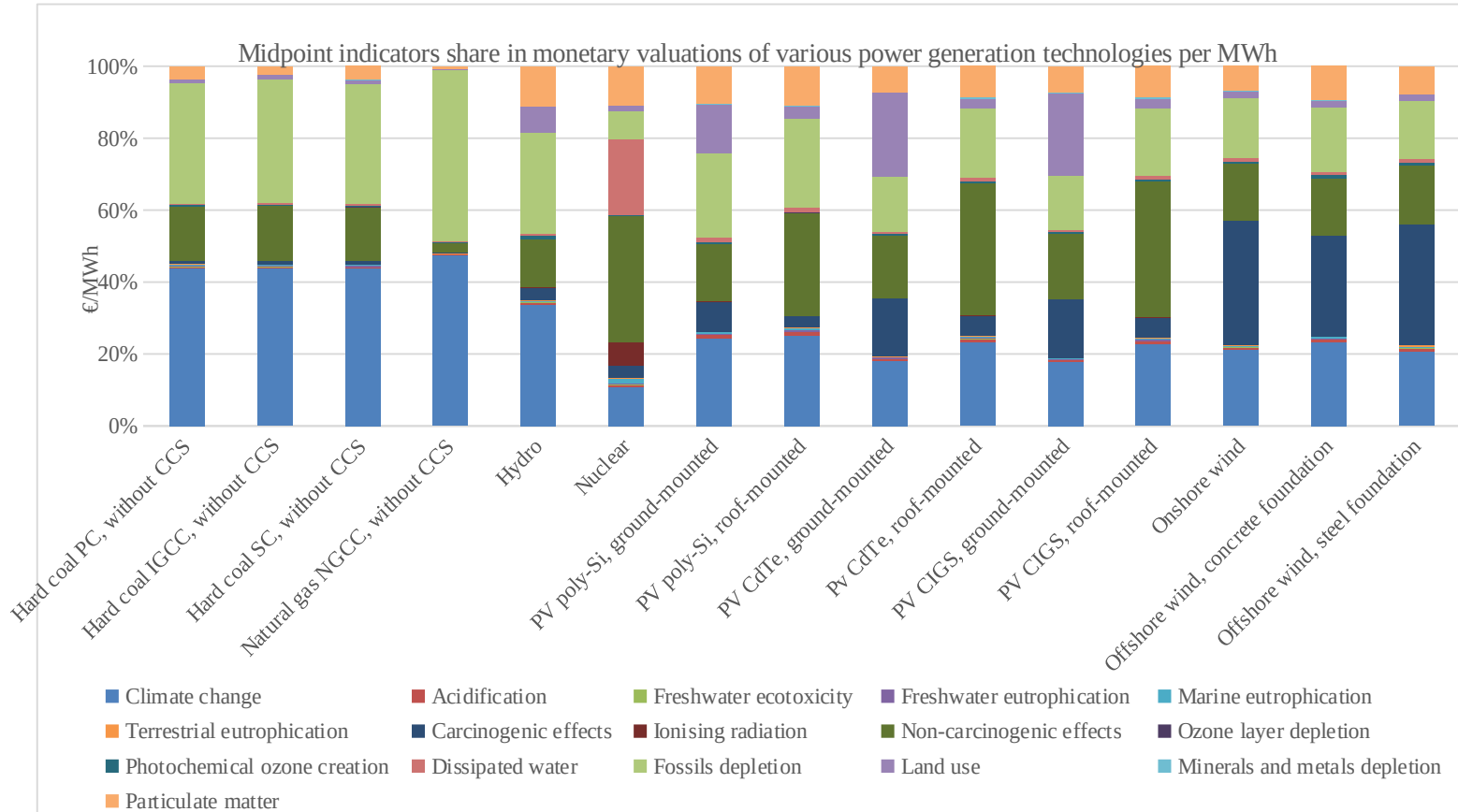
breakdown in percentage is detailed for each indicator in Fig. 2. Overall, climate change, fossils depletion (except for nuclear) and non-carcinogenic effects (mainly due to arsenic ion emissions [42]) are the largest contributors to environmental costs. As for nuclear, ionizing radiation contributes to roughly 6 % of the environmental costs with

⁵ More accurately, partially non-renewable since uranium can be reprocessed.

⁶ Which costs 12 €/MWh for both ground-mounted and roof-mounted configuration.

the same order of magnitude as particulate matter emissions (10 %, occurring mainly during the mining stage [44]) and fossils depletion (8 %, see above). With the strong caveat discussed above for water dissipation indicator, the related cost contributes 20 % of the environmental cost, thus amounting to almost 2 €/MWh.

FIG. 2. Midpoint indicators share in monetary valuations of various power generation technologies per MWh.
NB: NGCC = natural gas combined cycle

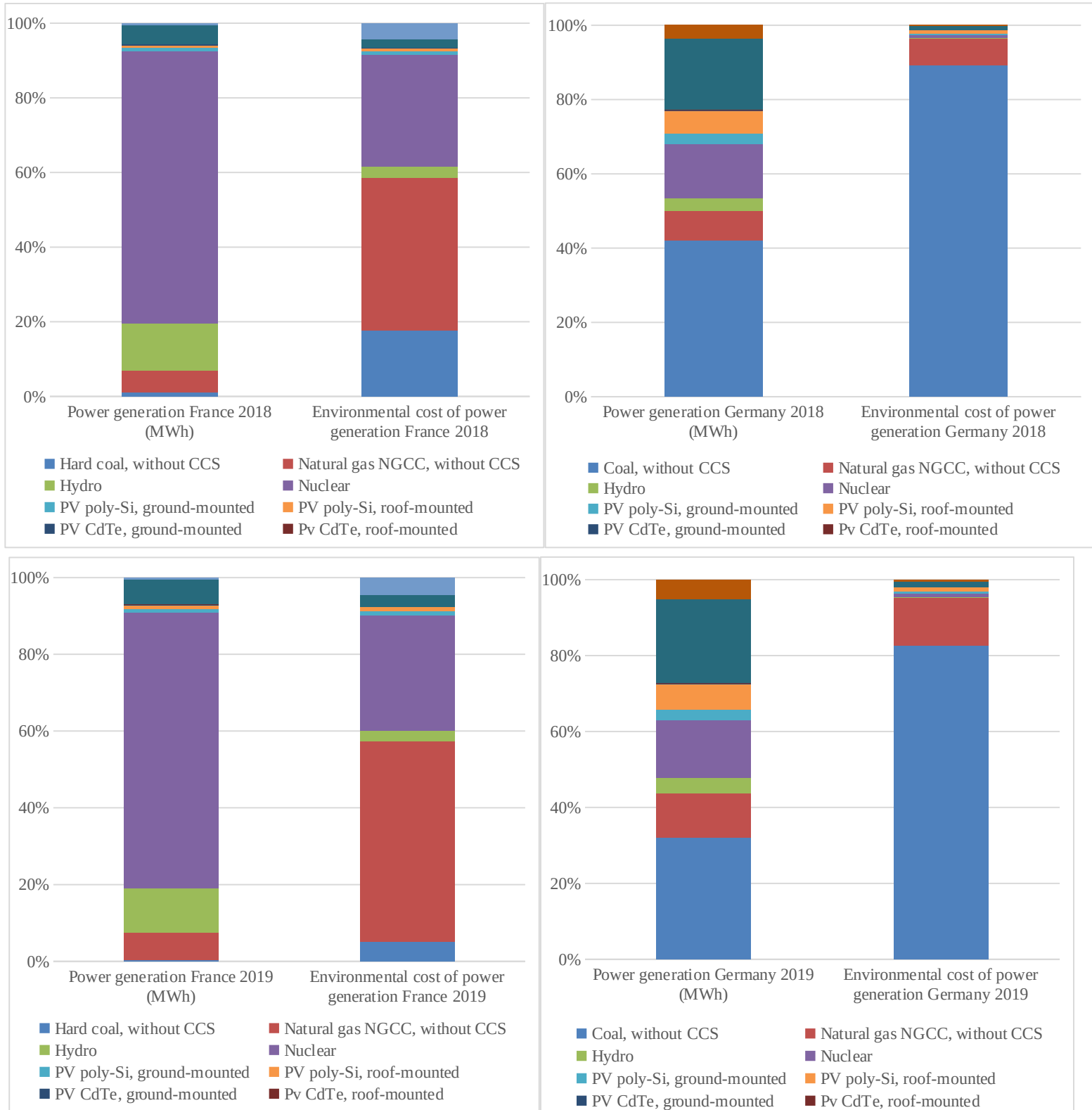


Overall, the environmental cost of power generation in France for the period 2018-2019 is about 23 €/MWh (Tab. 4.). Technological breakdown (Fig. 3.) shows that although accounting for 5 % of power production, natural gas contributes 40 % of national environmental cost. On the contrary, nuclear energy contributes more than 65 % of national power output but represents only 30 % of national environmental cost of generating one unit of electricity. We conclude as well that while generating 40 % of power to the global German national grid, hard and brown coal represent 80 % of its environmental cost. These results arise from a comparison of the environmental cost per technology with the average environmental cost for the entire grid within the perimeter of domestic generation (minus other renewables outside the scope of the study, see Materials and methods part). We can then sort the technologies between those that are *environmentally beneficial* in the current configuration of the power system (nuclear, renewables and, to some extent, natural gas for the German grid) and those that are not (coal, and natural gas for the French grid). Were those costs internalised and priced on wholesale electricity markets on a pay-as-clear logic, we would expect a phase-out of environmentally non-beneficial technologies to the extent of environmentally beneficial technologies capacities, following mainstream economics [53-54].

TABLE 4. Overall environmental cost of power generation for France and Germany in 2018

	2018	2019
France	23,6 €/MWh	23,1 €/MWh
Germany	188 €/MWh	155 €/MWh

FIG. 3. Share in power generation and contribution of generation technologies to environmental cost in France (left) and Germany (right) in 2018 (above) and in 2019 (below).



4. DISCUSSION FOR FURTHER WORK

It should first be mentioned that when using imported⁷ results from external LCA, as it is done here, if the theoretical load factor used in hypothesis is higher than the actual one (the one from hourly output data), we underestimate impacts of generation, and vice versa. In addition, aggregation of complex phenomena occurring over different places⁸ and times is implicit when assessing the environmental cost of production as all impacts are assigned to electricity generation. Any aggregation of the complexity will reduce the dimension of the problem, this is inevitable. However, we would like to carry on this work further, if only to suggest ways to clarify environmental issues at stake with the role of nuclear energy in electricity mix in transition. Precisely, we hope to tackle three fundamental matters: *first*, what're the environmental consequences of nuclear power plant closures (data used for the model cover both Fessenheim and Philippsburg closures). *Secondly*, how environmental impact seasonality [34-35] varies between a mix based on a strong nuclear fleet as base load technology. *Third*, what level of investment in high-performance operating nuclear fleet is rationale, should environmental cost be internalized [56]. In conclusion, beyond the question of the impact of the future net-zero stabilized power fleet, we intend to study what the least environmentally damaging short-term pathway to a sustainable mix is, based on an economic assessment of the environmental footprint of electricity generation.

5. FURTHER INFORMATION

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⁷ At this stage, we have not planned to compute our own LCA for the various technologies involved in this work.

⁸ In an extended review of different monetization methods in LCA [55], researchers conclude that “the most influential criterion [is] geographical reference”.

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